

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

Prepared by: Pranjal Sorte

Prepared for: Head of Supply Chain & Chief Financial Officer

Classification: Internal — Business Intelligence

Executive Summary

This report presents the findings of a four-year sales intelligence analysis conducted on retail transaction data from 2014 to 2018. Using advanced forecasting and machine learning techniques, we developed a system capable of predicting monthly sales with measurable accuracy, detecting unusual sales events before they escalate into revenue surprises, and categorizing the product portfolio into distinct demand groups that require different stocking strategies.

The key takeaway for the business is straightforward: the top revenue-generating product categories continue to grow, the November–December period remains the highest-risk and highest-reward season, and a significant portion of the product catalog is either declining or highly unpredictable, requiring immediate inventory rationalization.

1. Key Findings from Data Exploration and Analysis

Revenue Performance

- Technology is the highest-revenue category, generating approximately 36% of total revenue, followed by Furniture (32%) and Office Supplies (31%).
- Technology also carries the highest average order value, meaning fewer orders drive more revenue, creating a concentration risk.
- The West region is the largest revenue contributor, accounting for roughly 32% of total sales, with consistent year-over-year growth across all four years.
- Average shipping time across the business is 3.9 days, but varies by region. The South region averages 4.2 days, while the West averages 3.6 days, affecting customer satisfaction and repeat purchase rates.

Seasonality Pattern

Sales follow a strong seasonal pattern with two annual peaks:

1. September–October: Back-to-school and corporate purchasing season.
2. November–December: Holiday season, consistently generating 20–35% more revenue than average months.

The lowest revenue months are January and February, creating a predictable post-holiday slowdown that should inform inventory clearance strategies.

Time Series Signal

Time-series decomposition confirms that the business is in a steady growth phase. The underlying trend increased by approximately 22% over four years. Seasonal fluctuations exist but are not the primary growth driver, indicating healthy long-term business growth.

2. Three-Month Sales Forecast (August–October 2018)

Based on the best-performing forecasting model, the next three months are projected as follows:

Month	Forecasted Sales	Confidence Range	Notes
Month +1	~\$57,952	±\$13,847	End-of-summer transition period
Month +2	~\$62,067	±\$13,847	Back-to-school peak begins
Month +3	~\$65,021	±\$13,847	Q4 ramp-up before the holiday season

Plain Language Interpretation:

Sales are expected to remain stable in the first month and then increase meaningfully in Months 2 and 3 as the holiday season approaches. The model's average prediction error (MAE) on unseen data was approximately \$13,847 , meaning forecasts should remain accurate within that range under normal business conditions.

Note: Exact forecast values are generated from the notebook output file.

3. Top Three Anomalies Detected and Their Likely Causes

Anomaly 1: November Sales Spike (High Confidence)

Sales Level: Approximately 150–200% above the weekly average.

Detection Method: Flagged by both Isolation Forest and Z-Score methods.

Likely Cause: Black Friday and early holiday promotions.

Business Action: Maintain six to eight weeks of safety stock for top-selling Technology and Office Supplies products by early October.

Anomaly 2: January Sales Decline

Sales Level: Approximately 40–60% below the weekly average.

Detection Method: Flagged by the Z-Score method.

Likely Cause: Post-holiday reduction in consumer and business spending.

Business Action: Use this period for inventory counting, supplier negotiations, and clearance promotions. Delay major replenishment orders until February.

Anomaly 3: September Mid-Month Spike

Sales Level: Approximately 130% above the weekly average.

Detection Method: Flagged by Isolation Forest.

Likely Cause: Back-to-school corporate purchasing and bulk orders.

Business Action: Increase Office Supplies inventory before late August and introduce volume discounts to secure corporate orders.

4. Product Demand Segmentation and Stocking Strategy

Machine learning clustering identified four major demand groups across the product catalog.

Cluster 1: High Volume and Stable Demand

Example Products: Phones, Chairs, Storage, Binders.

Characteristics:

- High revenue generation.
- Consistent monthly demand.
- Low sales volatility.

Recommended Strategy:

Maintain automated reorder systems and keep four to six weeks of inventory coverage at all times.

Cluster 2: Growing Demand

Example Products: Machines, Accessories, Copiers.

Characteristics:

- More than 20% year-over-year growth.
- Increasing order volumes.

Recommended Strategy:

Increase purchase quantities by 15–20% every quarter and establish relationships with secondary suppliers.

Cluster 3: High Volatility

Example Products: Art, Envelopes, Fasteners.

Characteristics:

- High demand fluctuations.

- Moderate average sales.

Recommended Strategy:

Maintain 1.5–2 times normal safety stock and review inventory weekly.

Cluster 4: Declining Demand

Example Products: Tables and Bookcases.

Characteristics:

- Negative growth trends.
- Declining order frequency.

Recommended Strategy:

Reduce replenishment orders, launch markdown campaigns, and evaluate discontinuation of low-performing products.

5. Business Recommendations**Recommendation 1: Prepare Holiday Inventory by Early October****Supporting Data:**

Sales increase by 150–200% during November and December every year.

Action Plan:

- Order Technology and Office Supplies inventory during the first week of October.
- Order Furniture inventory six weeks before the holiday period.

Expected Impact:

Reduce stock shortages and recover an estimated 5–8% of peak-season revenue.

Recommendation 2: Rationalize the Bottom 20% of Products**Supporting Data:**

Clustering analysis identified several product categories with declining sales.

Action Plan:

- Conduct a product rationalization review.
- Run clearance campaigns for low-demand products.
- Reallocate warehouse space and investment toward growing categories.

Expected Impact:

Free warehouse capital and improve returns by focusing on higher-growth products.

Recommendation 3: Deploy Automated Anomaly Alerts

Supporting Data:

The anomaly detection system identified unusual demand patterns one to three weeks before they appeared in monthly reports.

Action Plan:

Integrate anomaly detection into weekly reporting systems. Trigger automatic email or Slack notifications whenever weekly sales exceed a Z-score threshold of 2.

Expected Impact:

Reduce emergency procurement costs and improve supply-chain responsiveness.

6. Risks and Limitations

The primary limitation of this system is that it relies entirely on historical patterns. The model cannot anticipate unprecedented events such as:

- New product launches.
- Competitor activity.
- Supply chain disruptions.
- Economic shocks.
- Inflation or pandemics.

In these situations, forecast accuracy may decrease until the model is retrained using updated data.

Therefore, the forecasting system should be treated as a decision-support tool rather than a replacement for human judgment. Procurement and sales teams should review forecasts regularly and incorporate external business knowledge when necessary.

We recommend retraining the forecasting models every quarter to ensure that they continue to reflect current market conditions.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib

- Statsmodels
- Prophet
- XGBoost
- Scikit-learn
- Isolation Forest
- K-Means Clustering